

Project 61: Speech-to-Text Converter

Goal: Build a tool that takes audio input (live or file) and converts it into accurate, readable text.

Methods

- **Live Microphone Input:** Using `speech_recognition`
 - **Audio File Transcription:** Using `Whisper` or `SpeechRecognition`
 - **Output:** Transcribed text saved to file or printed
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Step-by-Step Implementation

Step 1: Install Required Libraries

```
pip install openai-whisper speechrecognition pydub
```

`pydub` is needed to handle `.mp3` or convert formats.

Option A: Live Mic Transcription (Simple Setup)

```
import speech_recognition as sr

# Initialize recognizer
r = sr.Recognizer()

def mic_to_text():
    with sr.Microphone() as source:
```

```

print("🎤 Speak now...")
audio = r.listen(source)

try:
    text = r.recognize_google(audio)
    print("📝 Transcribed Text:\n", text)
except sr.UnknownValueError:
    print("⚠️ Could not understand the audio.")
except sr.RequestError as e:
    print("⚠️ Request failed; {0}".format(e))

```

Call it with:

```
mic_to_text()
```

Option B: Audio File Transcription with Whisper

```

import whisper

# Load Whisper model
model = whisper.load_model("base") # or use "small", "medium", "large"

def file_to_text(audio_path):
    print("🔍 Transcribing", audio_path)
    result = model.transcribe(audio_path)
    print("📝 Transcribed Text:\n", result["text"])
    return result["text"]

```

Call it with:

```
file_to_text("audio_sample.mp3")
```

Step 3: Save Transcription to a File

```
def save_transcription(text, output_file="transcript.txt"):
    with open(output_file, "w") as f:
        f.write(text)
    print(f"✅ Saved transcription to {output_file}")
```

Key Concepts

- **speech_recognition**: Works with online Google API (lightweight)
 - **whisper**: More accurate, multilingual, works offline
 - **pydub/ffmpeg**: Needed for audio conversion if using MP3s
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Project Summary

- **Input**: Microphone or audio file (MP3, WAV)
- **Model**: Google STT or OpenAI Whisper
- **Output**: Transcribed text (terminal or saved)
- **Use Case**: Podcasts, lectures, meetings, accessibility tools